

MAM03 – Advanced Data Analysis in Medicine

Week 2 assignment: renal transplantation and risk of graft failure

Laura Niesse (15942619)

Tyenca de Graf (15688097)

Methods

Data preparation and study description

The study cohort comprised 752 patients with renal failure who received a renal transplant. The primary outcome was time to graft rejection (gsurv, gstatus). An initial competing risk analysis was performed to determine the influence of patient death (pstatus) on the graft rejection analysis. It was determined that no patients in the cohort died with a functioning graft (gstatus = 0 and pstatus = 1). Therefore, patient death is not a competing risk for the analysis of graft rejection and a standard Cox proportional hazard model was selected. Alongside, a Fine-Gray model was also performed as a supplementary analysis to confirm that accounting for potential competing risks did not alter the findings.

Missing values within the covariate data were analyzed and subsequently imputed using MICE. For this analysis, a single imputed dataset was generated using 10 internal iterations.

Results

Univariable analysis

Table 1 (see Appendix A) shows the baseline characteristics of kidney transplantation patients (n=752), stratified by outcome (censored, graft rejection or death). The median age at transplantation was 45 years with no significant difference between groups. Delayed graft function (dggf) occurred in about 20% of the patients and was similarly distributed across outcomes.

The number of acute rejection treatments differed significantly ($p < 0.001$). Patients who experienced graft rejection logically showed more rejection episodes. Additionally, the median creatinine at six weeks was higher among patients who had death as outcome compared to those censored. Diastolic bloodpressure and PRA levels were similar between groups.

Protein in urine at six weeks was more common in the outcome death ($p = 0.009$). The distribution of shared CREGs showed no significant differences. Overall, most baseline variables were similar except for creatinine and protein urine.

To evaluate the association of each risk factor with the risk of graft rejection, univariable Cox regression analysis was performed. Cox regression analysis was used in order to perform univariable regression.

For the categorical variable uprotein, '>+' was dropped to serve as the reference category. This was done to avoid multicollinearity (perfect correlation between predictor variables), which can cause issues in regression models. The coefficients of the included dummy variables were then interpreted relative to this dropped reference category.

The results, presented in Table , show that several factors had a statistically significant ($p < 0.05$) association with the risk of graft rejection. Significant were patient age (acclft), number of cross-reactive groups (cregsh), delayed graft function (dggf_dggf), number of acute rejection treatments (aantalre) and urine protein level (uprotein).

Where coef = The log hazard ratio, $\exp(\text{coef})$ = the Hazard Ratio (HR).

covariate	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	p	-log2(p)
acclft	0.0127	1.0127	0.0055	0.0019	0.0234	1.0019	1.0237	0.0216	5.5346
creat	0.0021	1.0021	0.0007	0.0006	0.0035	1.0006	1.0035	0.0048	7.7122
predias	0.0066	1.0067	0.0053	-0.0037	0.017	0.9963	1.0171	0.2082	2.2639
prac	0.0036	1.0036	0.0028	-0.002	0.0092	0.998	1.0093	0.2031	2.2999
cregsh	-0.1621	0.8504	0.0565	-0.2729	-0.0513	0.7612	0.95	0.0041	7.9198
dgf_dgf	-0.0916	0.9125	0.1688	-0.4224	0.2393	0.6554	1.2703	0.5875	0.7674
aantalre_1.0	0.4247	1.5291	0.1846	0.0629	0.7865	1.0649	2.1956	0.0214	5.5463
aantalre_2.0	0.6221	1.8628	0.1861	0.2573	0.9868	1.2935	2.6826	0.0008	10.2362
aantalre_3.0	1.4009	4.0587	0.1918	1.025	1.7768	2.787	5.9107	0.0	41.7033
uprotein_geen	-1.3054	0.2711	0.2256	-1.7476	-0.8633	0.1742	0.4218	0.0	27.0509
uprotein_s p/+	-0.7938	0.4521	0.2101	-1.2055	-0.3821	0.2995	0.6825	0.0002	12.6305

Table 2 - results univariable analysis

Multivariable Analysis & Model validation

A multivariable Cox regression model was subsequently fit, including all 8 risk factors. Prior to interpretation, the key assumptions of this model were systematically evaluated.

The Variance Inflation Factor (VIF) was calculated for all predictors in the model to detect multicollinearity. High multicollinearity indicates that one or more predictors are strongly correlated with other predictors in the model. This can cause instability in coefficient estimates or even prevent convergence. VIF values were interpreted according to the following guidelines:

- 1–5: Desired range; low multicollinearity
- 5–10: Moderate multicollinearity; further investigation is required
- 10: High multicollinearity; consider removing or combining variables

As shown in Appendix B, all VIF values were below the threshold of 5. Therefore, we do not have problematic multicollinearity among our predictors in the Cox regression model.

The assumption of a linear relationship between continuous predictors and the log-hazards was tested. A visual inspection of the Martingale residual plots showed a random scatter around zero, supporting linearity assumption (see Appendix C). In addition to a visual check, a Cox model for each continuous variable was fitted, comparing a model with a linear term to a more flexible model using splines (4 degrees of freedom). This tells us if the linear assumption for this variable is sufficient, or if we need a flexible shape (like splines). Lastly, a Likelihood Ratio Test (LRT) was performed to compare results. The categorical variables are excluded because they will not provide us with statistically reliable insights.

The results of the LRT p-values can be found in Appendix D. The plots visualizing the partial effect on the Log-hazard ratio and the partial effect on the Log-hazard ratio with splines are presented in Appendix E. For all tested continuous variables, the p-value of the LRT was greater than 0.05. This suggests that a linear model is sufficient to describe the relationship and we do not reject the null hypothesis. This means there is no statistically significant evidence against the linear assumption.

The proportional hazards (PH) assumption was evaluated using scaled Schoenfeld residual plots (see Figure 1) and the corresponding statistical tests (see Appendix F). The null hypothesis states that the covariate's effect is constant over time. This assumption was violated for two predictors, namely age ($p < 0.005$) and aantalre3.0 ($p = 0.02$). This suggests that the effect of patient age and the effect of having three rejection episodes on the risk of graft rejection are not constant over time. The assumption was met for all other covariates.

($p > 0.05$). This violation represents an important finding, and the non-proportional variables were appropriately addressed before constructing the final model. The remaining Schoenfeld residual plots are presented in Appendix G.

The relationship between the risk factors and the time-to-event outcome was modeled using Cox proportional hazards regression. This model is commonly used for this type of data because it does not require us to specify the exact shape of the baseline hazard over time. It inherently uses a log-link function for the hazard and directly yields interpretable hazard ratios. Alternative link functions, such as the logistic or log-log, were not tested by us as the primary focus of this analysis is on estimating hazard ratios over time rather than absolute event probabilities.

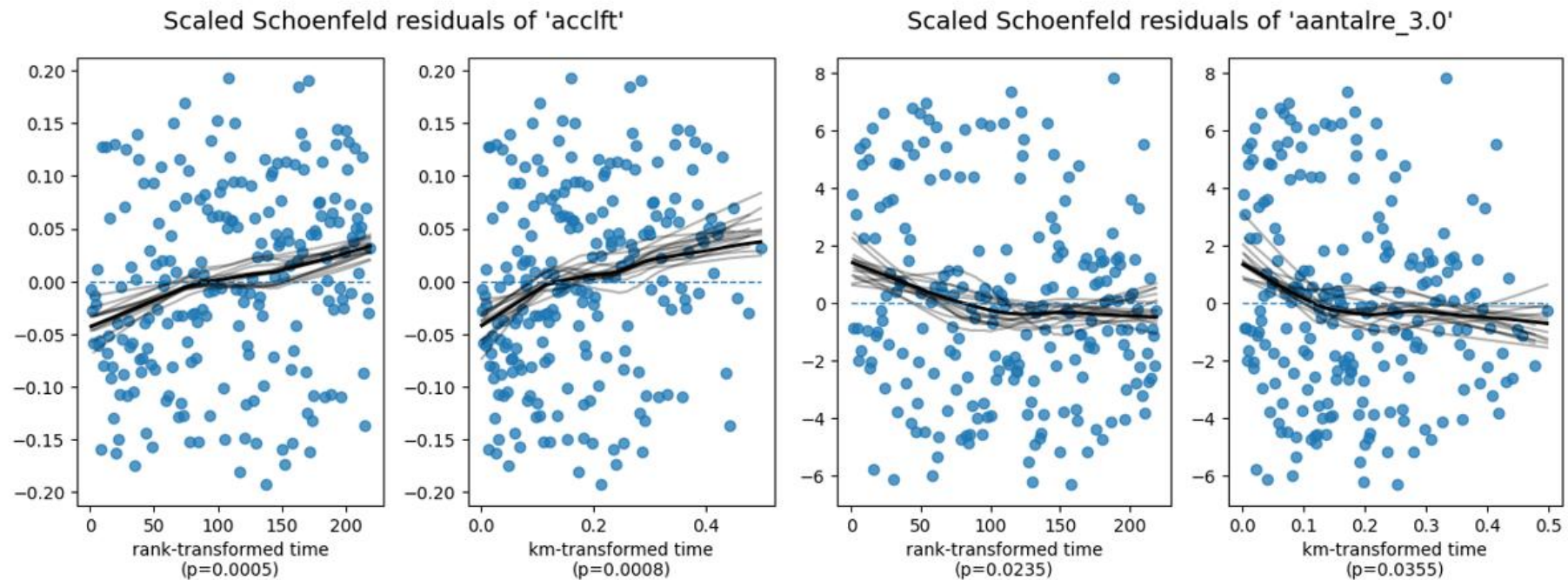


Figure 1 - Schoenfeld residuals where $p \leq 0.05$

Violated assumptions

As established previously, the variables `acclft` and `aantal_3.0` are not constant over time and therefore violate the proportional hazards assumption. A landmark Cox model was used to partition the follow-up time to graft rejection into different time intervals.

A multivariable Cox regression was then fitted within each interval to adjust for the influence of other covariates.

For `acclft`, it can be concluded that during the first three years after transplantation (0–36 months), the effect of 'acclft' on graft rejection was small and non-significant ($p > 0.16$). Between 36 and 48 months, the predictor became statistically significant ($p < 0.05$), indicating that older recipients had a slightly higher risk of graft rejection. After 60 months, this effect remained highly significant and slightly stronger ($p < 0.001$).

For `Aantalre_3.0`, the first three years had the strongest effect, as indicated by the p-values ranging from $p \leq 0.001$ to $p \leq 0.01$. After these three years the effect became statistically non-significant, suggesting that the long-term rejection risk for patients with a history of three or more rejection treatments decreases over time. The full results are presented in Appendix H.

Final model

After verifying model assumptions and addressing the identified violations, a final multivariable Cox proportional hazards model with time-varying covariates was fitted (see Table 3), along with a Fine–Gray competing-risks model to provide a more complete overview. The proportional hazards assumption was violated for age at transplantation (`acclft`) and the number of acute rejection treatments (`aantalre_3.0`). Therefore, these variables were modeled with coefficients that could vary across predefined graft-survival intervals, while other predictors were treated as time-fixed.

The predictor proteinuria (`uprotein`) was strongly associated with a lower rejection risk. Compared to the reference category ($>+$), patients with no protein in urine had a 52% lower hazard of rejection (HR = 0.48, 95% CI: 0.33–0.71, $p < 0.001$), and those with trace amounts (`sp/+`) had a 32% lower hazard (HR = 0.68, 95% CI: 0.47–0.97, $p = 0.034$). Delayed graft function (`dgf_dgf`) showed a potentially protective effect (HR = 0.73, 95% CI: 0.52–1.01, $p = 0.057$), while a higher number of cross-reactive groups (`cregsh`) was significantly associated with a lower hazard of rejection (HR = 0.89, 95% CI: 0.80–0.99, $p = 0.028$).

To account for the competing risk of patient death, a Fine–Gray hazard model was fitted using the same covariates as the Cox model (see Appendix I). In this model, all covariate effects were treated as time-fixed, providing an average estimate of how age at transplantation (acclft) and the number of acute rejection treatments (aantalre) are associated with the cumulative incidence of graft rejection. The results largely confirmed the findings from the Cox regression: both acclft and aantalre were strong predictors of graft rejection, while proteinuria, delayed graft function, and cross-reactive groups (cregsh) were protective.

covariate	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	p	-log2(p)
creat	0.0006	1.0006	0.0008	-0.0011	0.0023	0.9989	1.0023	0.4901	1.0288
predias	0.0066	1.0066	0.0052	-0.0035	0.0167	0.9965	1.0169	0.2025	2.3039
prac	0.0037	1.0037	0.0028	-0.0019	0.0093	0.9981	1.0093	0.1955	2.3545
cregsh	-0.1184	0.8883	0.054	-0.2242	-0.0126	0.7991	0.9874	0.0282	5.1462
dgf_dgf	-0.3189	0.727	0.1675	-0.6472	0.0095	0.5235	1.0096	0.057	4.1325
aantalre_1.0	0.191	1.2104	0.1613	-0.1251	0.507	0.8824	1.6604	0.2364	2.0806
aantalre_2.0	0.3255	1.3848	0.1688	-0.0054	0.6564	0.9946	1.9279	0.0538	4.2151
uprotein_geen	-0.7249	0.4844	0.1936	-1.1043	-0.3455	0.3314	0.7079	0.0002	12.4357
uprotein_sp/+	-0.3877	0.6786	0.1826	-0.7457	-0.0297	0.4744	0.9707	0.0338	4.8882
acclft_x_12-24	-0.0033	0.9967	0.0073	-0.0177	0.0111	0.9825	1.0112	0.6539	0.6128
acclft_x_24-36	0.0002	1.0002	0.0079	-0.0154	0.0157	0.9847	1.0159	0.9836	0.0238
acclft_x_36-48	0.0106	1.0106	0.0084	-0.0058	0.0269	0.9942	1.0273	0.2065	2.2761
acclft_x_48-60	0.0028	1.0028	0.0094	-0.0155	0.0211	0.9846	1.0213	0.7656	0.3854
acclft_x_60-170	0.0217	1.0219	0.0076	0.0068	0.0366	1.0069	1.0373	0.0042	7.8839
aantalre_3.0_x_12-24	1.5211	4.5771	0.331	0.8722	2.1699	2.3923	8.7572	0.0	17.8169
aantalre_3.0_x_24-36	0.7221	2.0588	0.4514	-0.1626	1.6068	0.85	4.9868	0.1096	3.1892

aantalre_3.0_x_36-48	0.6359	1.8886	0.4854	-0.3155	1.5872	0.7295	4.8899	0.1902	2.3945
aantalre_3.0_x_48-60	0.6428	1.9018	0.6119	-0.5566	1.8422	0.5732	6.3104	0.2935	1.7684
aantalre_3.0_x_60-170	0.6242	1.8667	0.411	-0.1813	1.4297	0.8342	4.1772	0.1288	2.9567

Table 3 - results multivariable analysis

Model performance

Cox regression model

The predictive performance of the final Cox model and final Fine-Grey model were evaluated at several landmark time points (12, 24, 36, 48, and 60 months) by computing the Concordance Index (C-index) and Brier score.

The C-index measures the model's ability to correctly rank the risk of two randomly selected individuals, whereas the Brier Score measures the accuracy of predicted survival probabilities, with lower values indicating better model performance.

For the Cox model the Brier Scores increased steadily over time (see Appendix J). indicating a gradual rise in prediction error as the follow-up period progressed. The model exhibited its highest accuracy at 12 months (Brier Score = 0.0069). The error increased to 0.0398 at 60 months, which still represents a relatively low level of prediction error across the follow-up period. According to standard guidelines, a C-index between 0.60 and 0.70 represents modest discrimination, which is considered an acceptable level of predictive performance for clinical survival models. The calibration plot (see Appendix K) shows most points clustered at the right upper corner. It suggests that both the predicted and observed survival probabilities are very high, which is consistent with our data.

Fine & Gray

We were not able to successfully calculate the C-index and Brier score for the Fine & Gray model. This is because the available Python functions did not work for our competing risk models. An attempt was made in R, but did also not succeed. Therefore, only the model coefficients and subdistribution hazard ratios (SHRs) were reported.

Appendix A

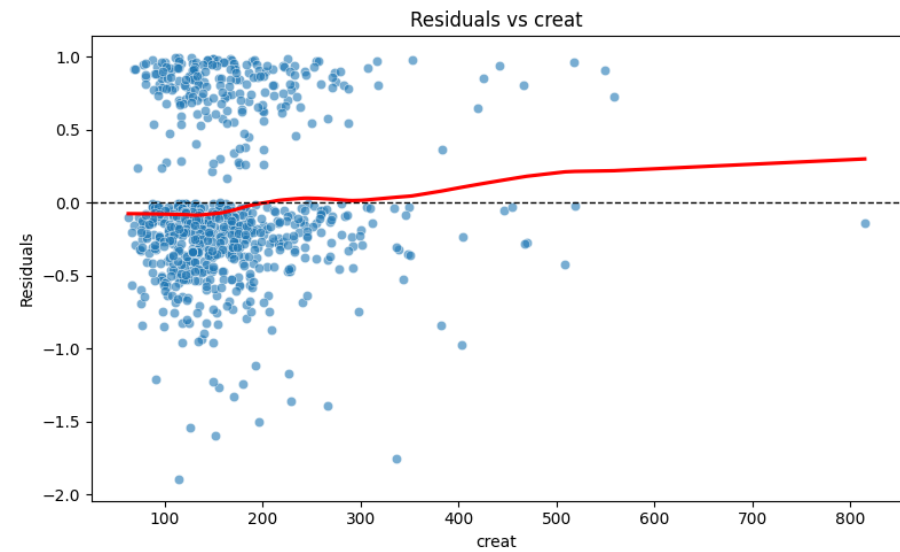
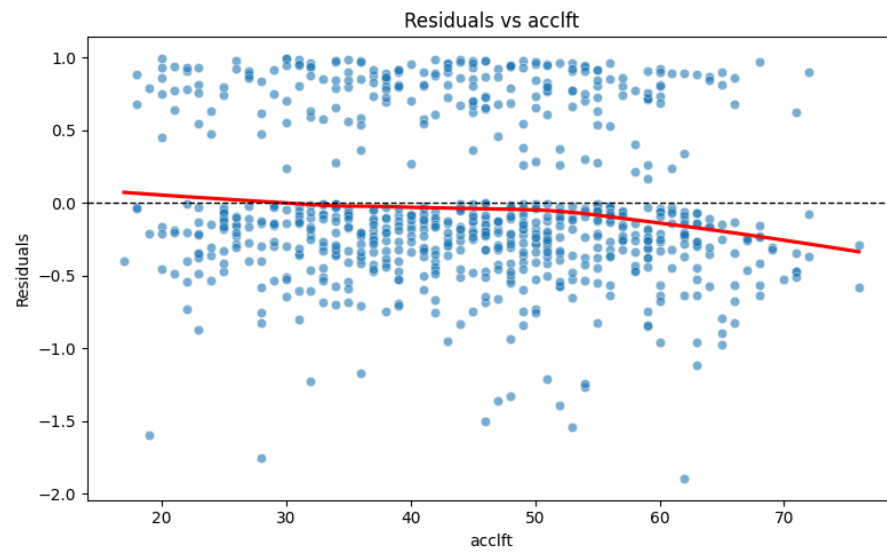
table 1. baseline vars by outcome (censored / reject / death)						
variable	N	Overall N = 752 (1%)	censored N = 533 (0.708776595744681%)	reject N = 219 (0.291223404255319%)	death N = 0 (0%)	p-value [†]
age tx (yrs), Median (Q1–Q3)	752	45.0 (35.0–54.0)	45.0 (35.0–54.0)	46.0 (35.0–56.0)	NA (NA–NA)	0.18
dgf, n (%)	738					0.93
0		583.0 (79.0%)	422.0 (79.2%)	161.0 (78.5%)	0.0 (NA%)	
dgf		155.0 (21.0%)	111.0 (20.8%)	44.0 (21.5%)	0.0 (NA%)	
Unknown		14	0	14	0	
acute rej tx (n), n (%)	752					<0.001
0		316.0 (42.0%)	248.0 (46.5%)	68.0 (31.1%)	0.0 (NA%)	
1		164.0 (21.8%)	112.0 (21.0%)	52.0 (23.7%)	0.0 (NA%)	
2		168.0 (22.3%)	117.0 (22.0%)	51.0 (23.3%)	0.0 (NA%)	
3		104.0 (13.8%)	56.0 (10.5%)	48.0 (21.9%)	0.0 (NA%)	
creat 6w (umol/L), Median (Q1–Q3)	751	148.0 (118.0–191.0)	144.0 (117.0–185.5)	160.0 (126.0–212.0)	NA (NA–NA)	0.003
Unknown		1	1	0	0	
dias bp pre-tx, Median (Q1–Q3)	748	90.0 (80.0–100.0)	90.0 (80.0–100.0)	90.0 (80.0–100.0)	NA (NA–NA)	0.37
Unknown		4	1	3	0	
pra (%), Median (Q1–Q3)	752	1.0 (0.0–8.0)	1.0 (0.0–7.0)	1.0 (0.0–9.0)	NA (NA–NA)	0.75
urine protein 6w, n (%)	734					0.009
geen		295.0 (40.2%)	226.0 (43.0%)	69.0 (33.2%)	0.0 (NA%)	
sp/+		401.0 (54.6%)	279.0 (53.0%)	122.0 (58.7%)	0.0 (NA%)	
> +		38.0 (5.2%)	21.0 (4.0%)	17.0 (8.2%)	0.0 (NA%)	
Unknown		18	7	11	0	
shared cregs (n), n (%)	735					0.19
1		6.0 (0.8%)	4.0 (0.8%)	2.0 (1.0%)	0.0 (NA%)	
2		12.0 (1.6%)	7.0 (1.3%)	5.0 (2.4%)	0.0 (NA%)	
3		121.0 (16.5%)	76.0 (14.4%)	45.0 (21.5%)	0.0 (NA%)	
4		225.0 (30.6%)	161.0 (30.6%)	64.0 (30.6%)	0.0 (NA%)	
5		215.0 (29.3%)	156.0 (29.7%)	59.0 (28.2%)	0.0 (NA%)	
6		109.0 (14.8%)	83.0 (15.8%)	26.0 (12.4%)	0.0 (NA%)	
7		41.0 (5.6%)	34.0 (6.5%)	7.0 (3.3%)	0.0 (NA%)	
8		6.0 (0.8%)	5.0 (1.0%)	1.0 (0.5%)	0.0 (NA%)	
Unknown		17	7	10	0	

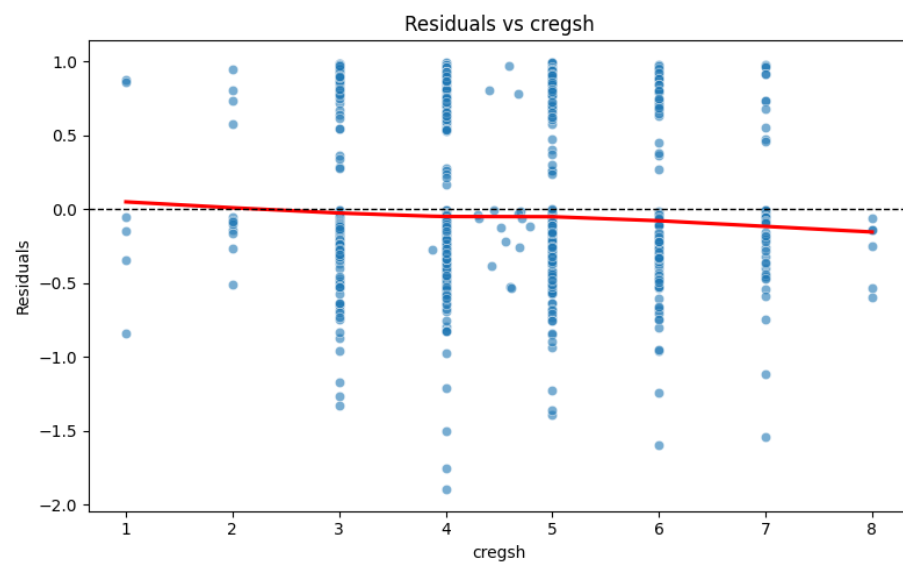
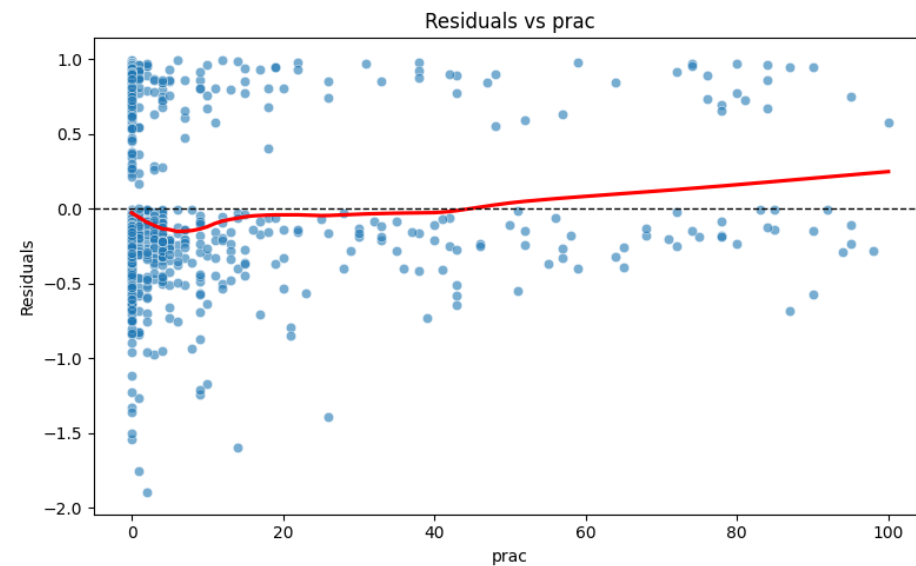
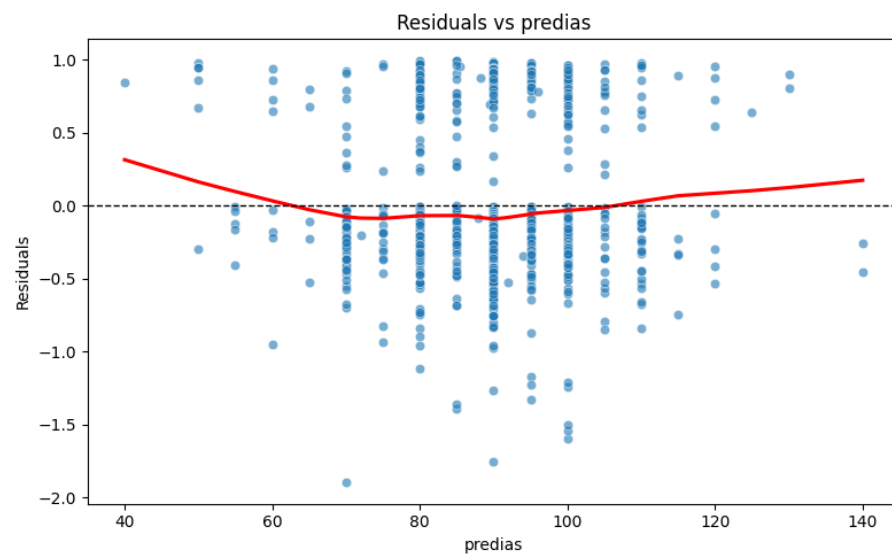
Appendix B

Index	Variable	VIF
0	acclft	1.0188
1	creat	1.3664
2	predias	1.0325
3	prac	1.0555
4	cregsh	1.0358
5	dgf_dgf	1.2084
6	aantalre_1.0	1.2175
7	aantalre_2.0	1.3090
8	aantalre_3.0	1.3676
9	uprotein_geen	4.0104
10	uprotein_sp/+	3.8498

Table 4 - results VIF

Appendix C



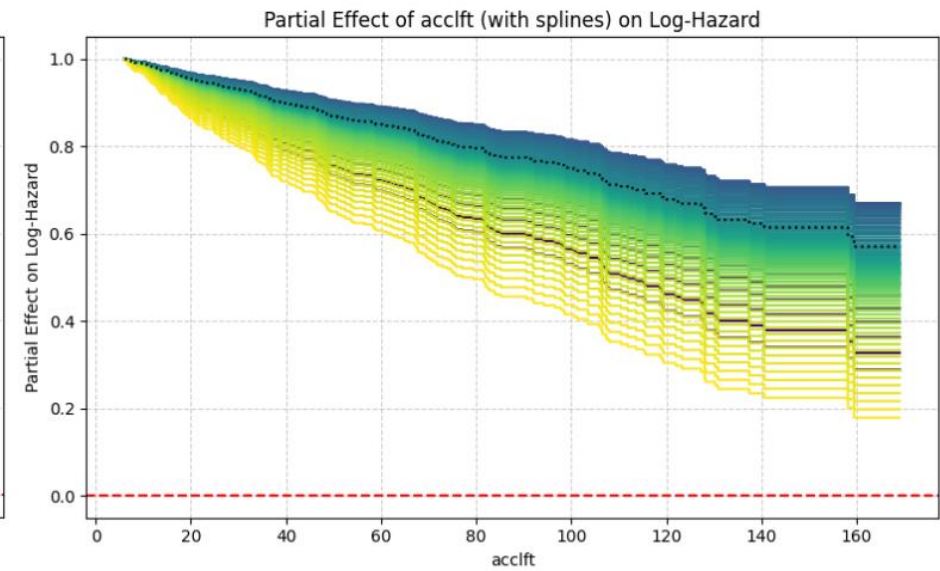
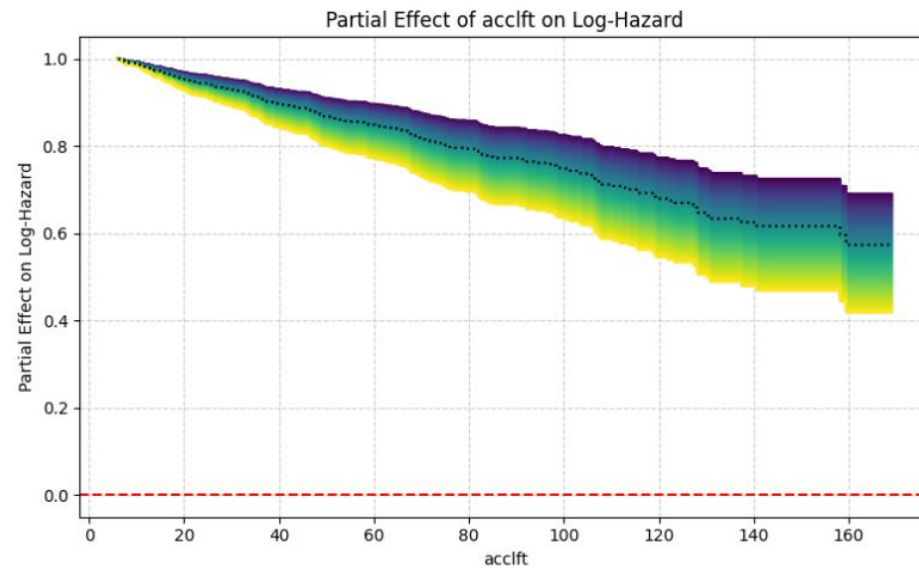


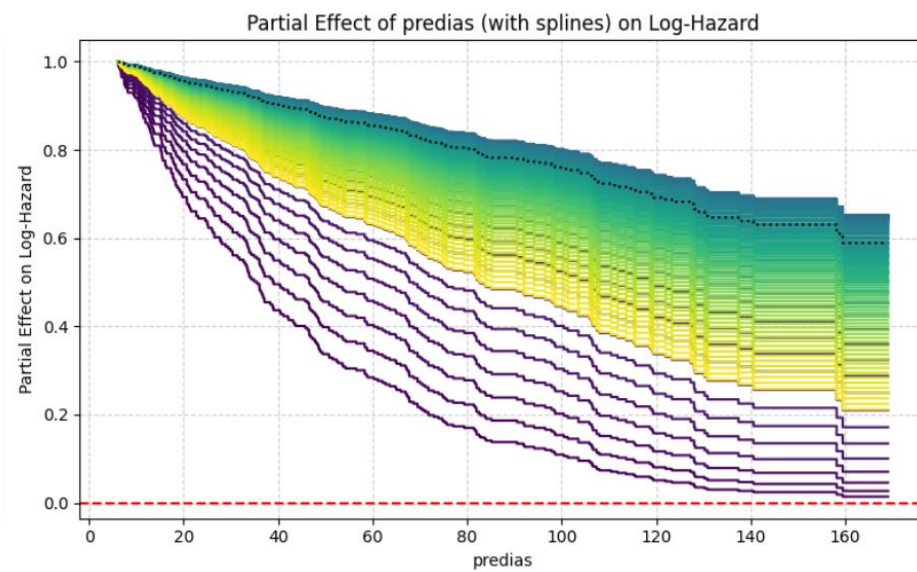
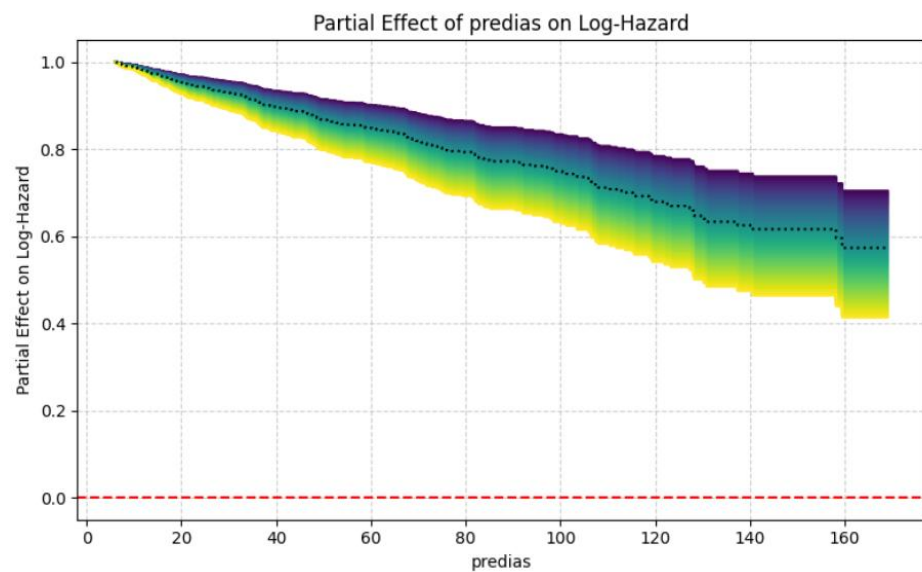
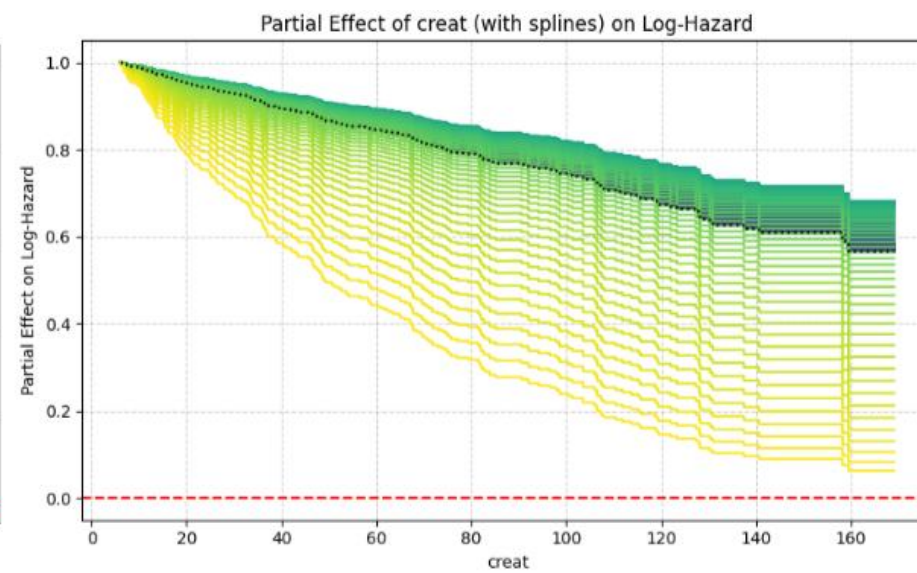
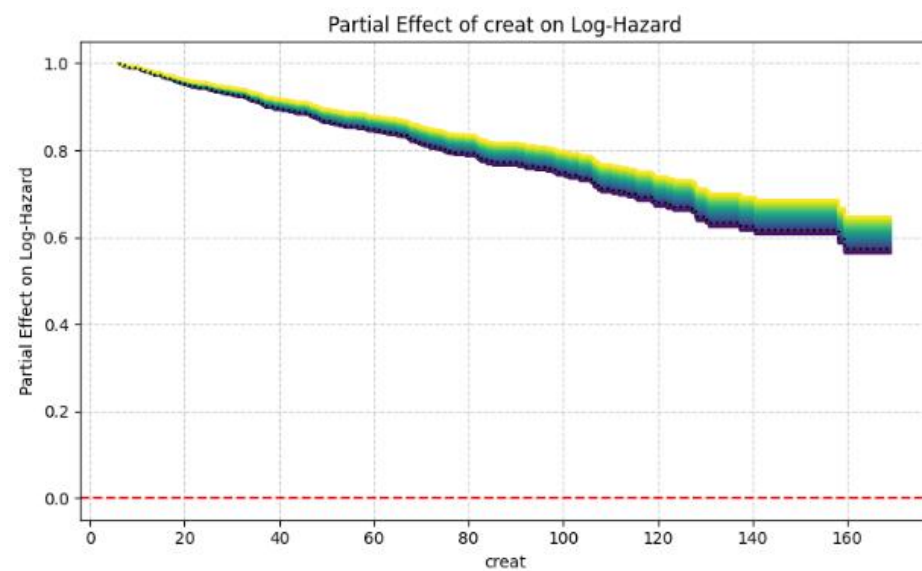
Appendix D

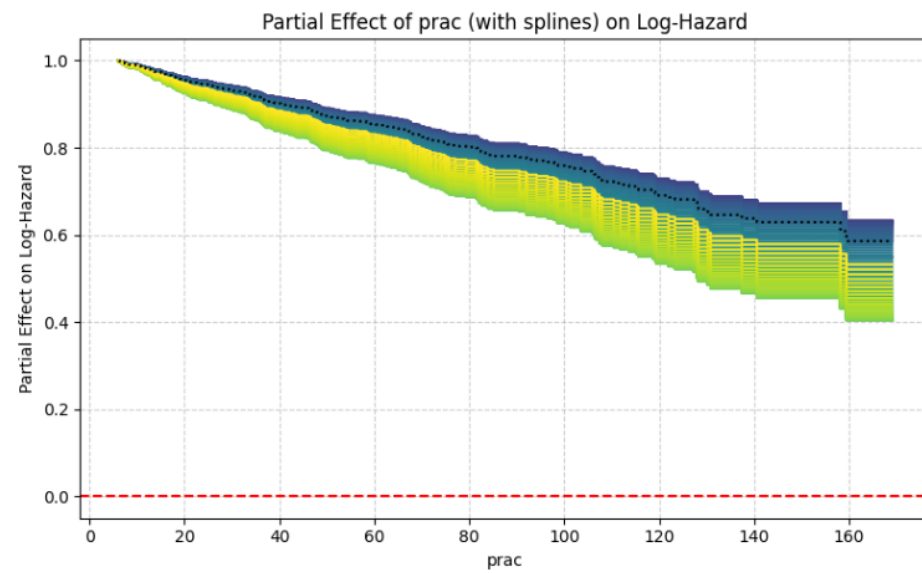
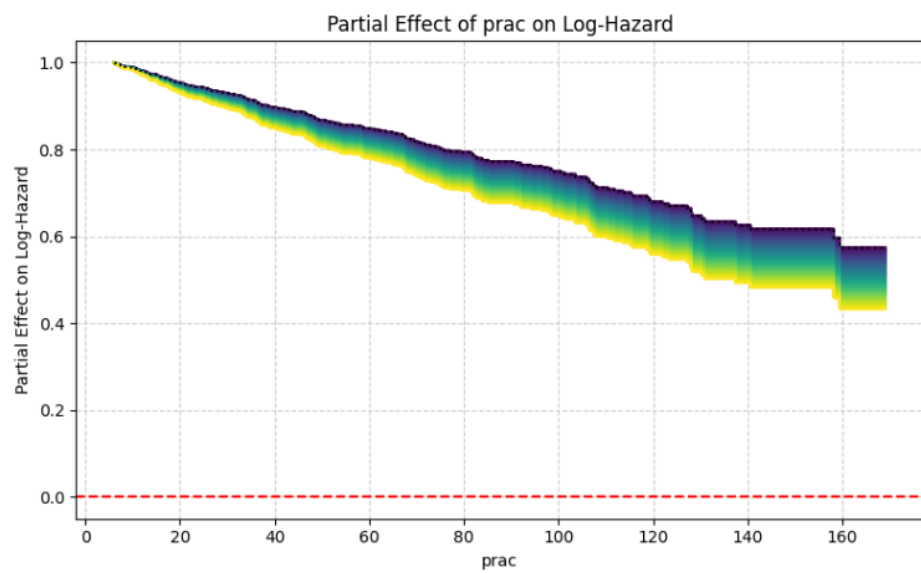
Covariate	p-value
Acclft	0.095466
Creat	0.472457
Predias	5.4749
Prac	3.1548
Cregsh	0.7205

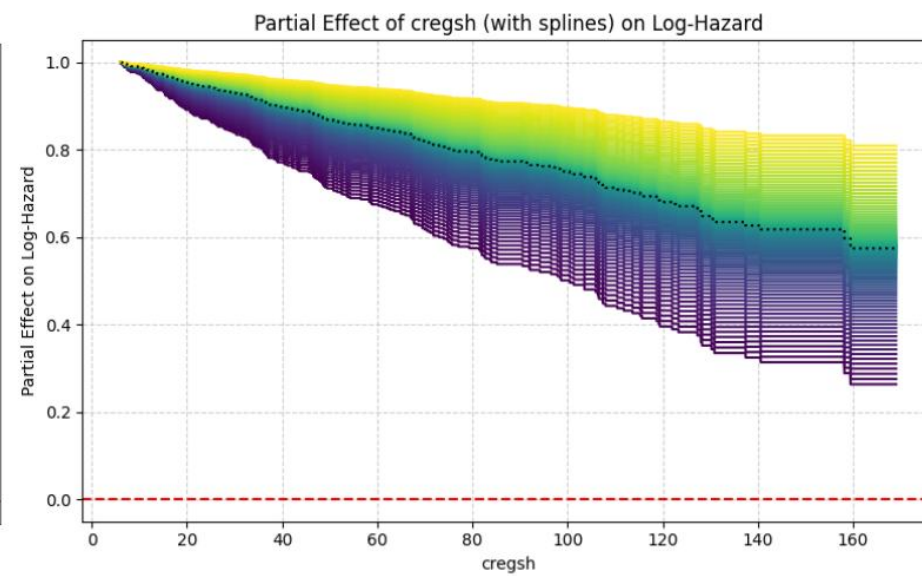
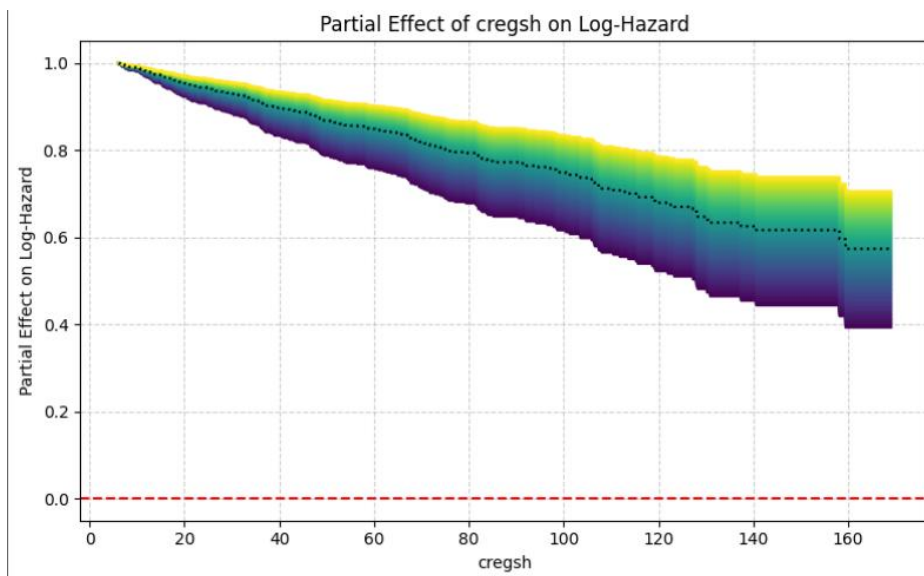
Table 5 - results LRT

Appendix E









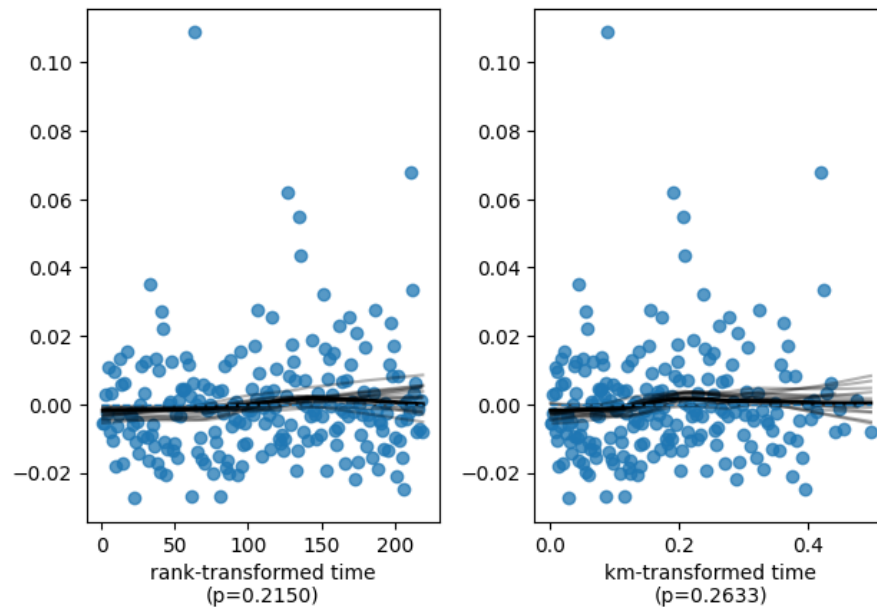
Appendix F

covariate		p	-log2(p)
aantalre_1.0	km	0.47	1.10
	rank	0.47	1.10
aantalre_2.0	km	0.11	3.21
	rank	0.12	3.06
aantalre_3.0	km	0.04	4.82
	rank	0.02	5.41
acclft	km	<0.005	10.32
	rank	<0.005	11.05
creat	km	0.26	1.93
	rank	0.21	2.22
cregsh	km	0.29	1.80
	rank	0.31	1.71
dgf_dgf	km	0.19	2.37
	rank	0.11	3.18
prac	km	0.27	1.91
	rank	0.13	2.96
predias	km	0.48	1.04
	rank	0.41	1.27
uprotein_geen	km	0.25	1.97
	rank	0.15	2.70
uprotein_sp/+	km	0.57	0.81
	rank	0.45	1.16

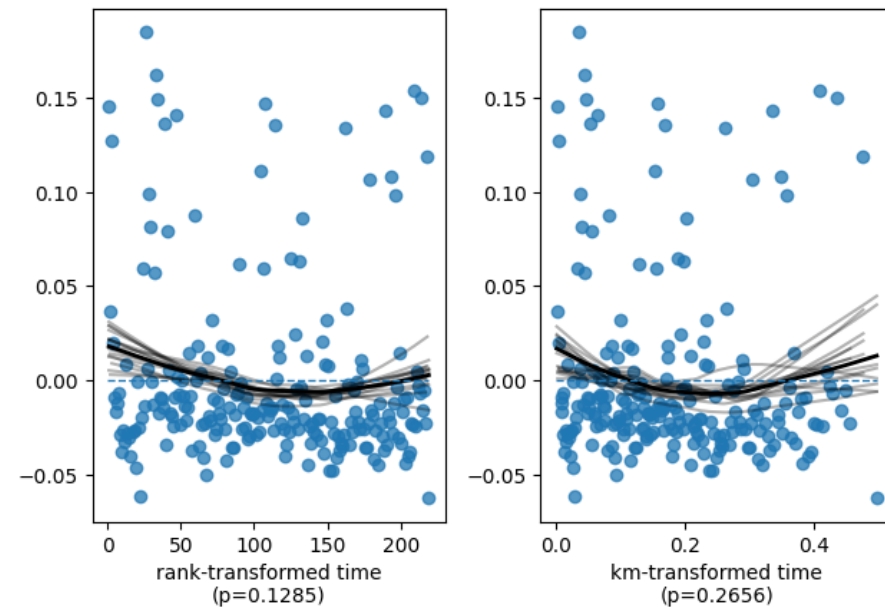
Table 5 - results of rank-transformed time and Kaplan–Meier transformed time.

Appendix G

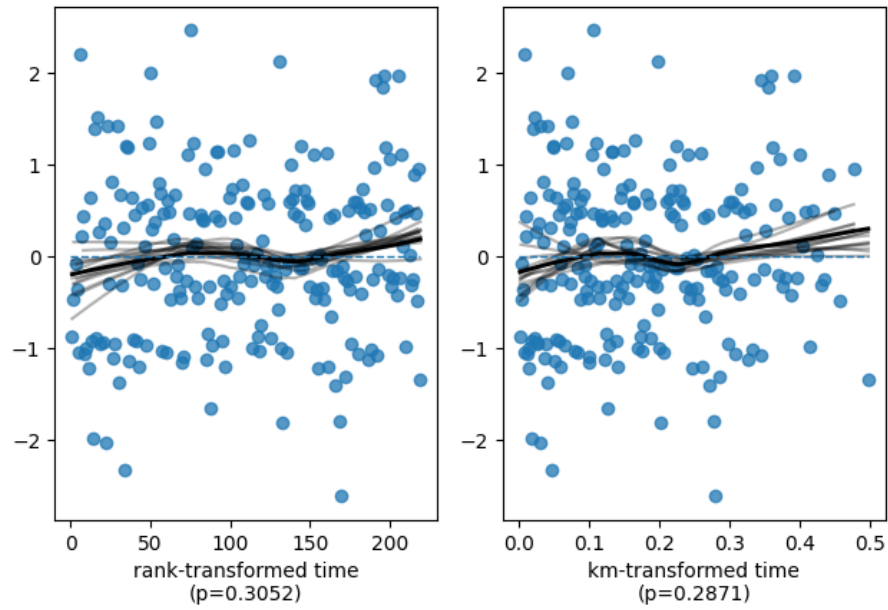
Scaled Schoenfeld residuals of 'creat'



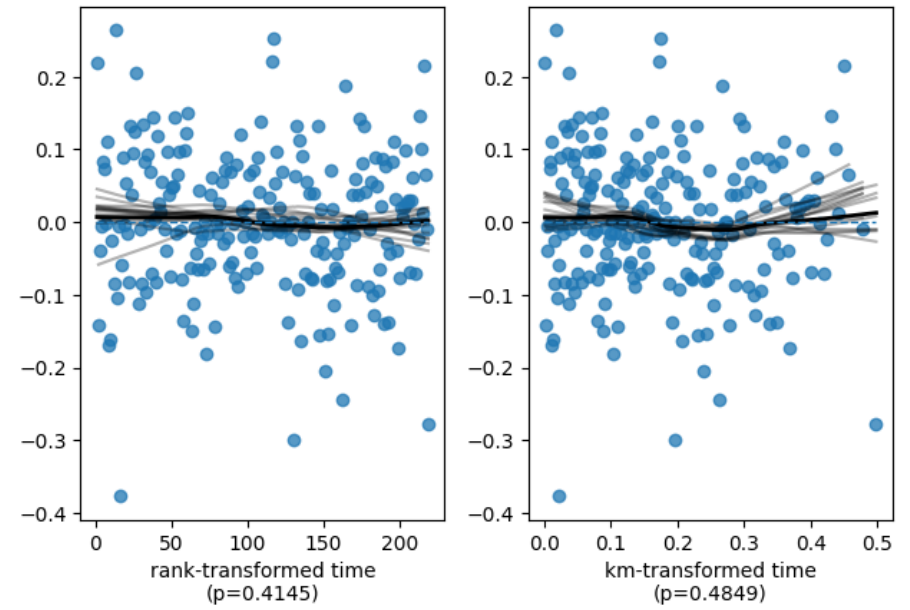
Scaled Schoenfeld residuals of 'prac'



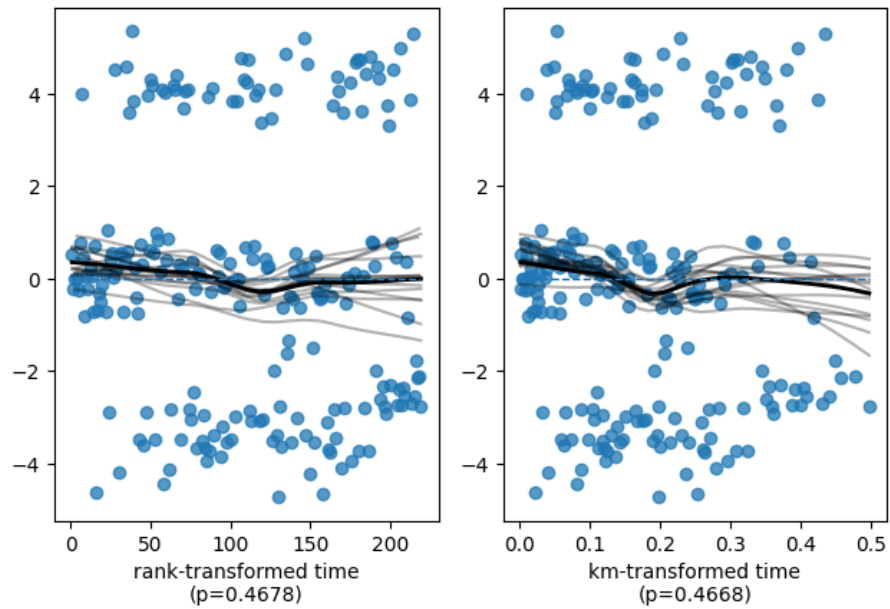
Scaled Schoenfeld residuals of 'cregsh'



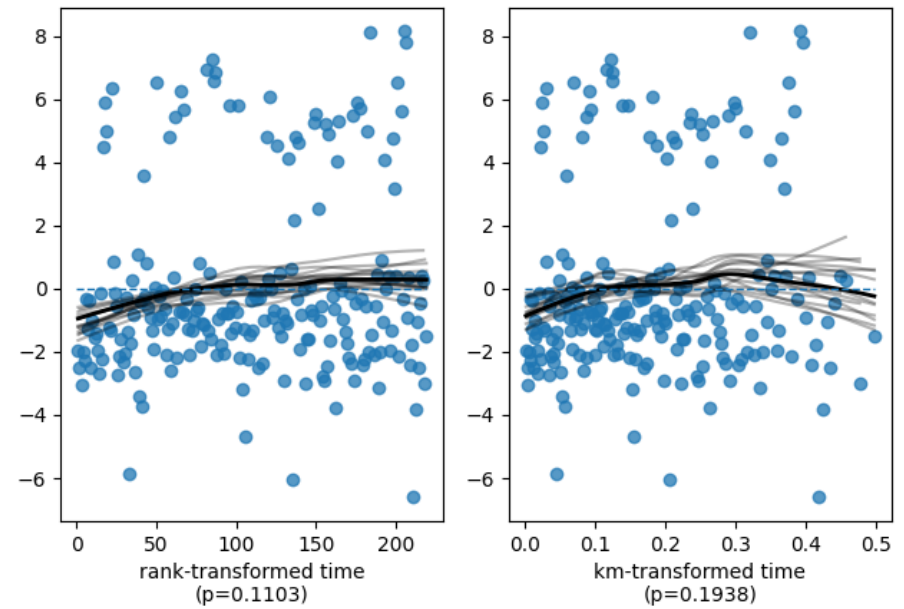
Scaled Schoenfeld residuals of 'predias'



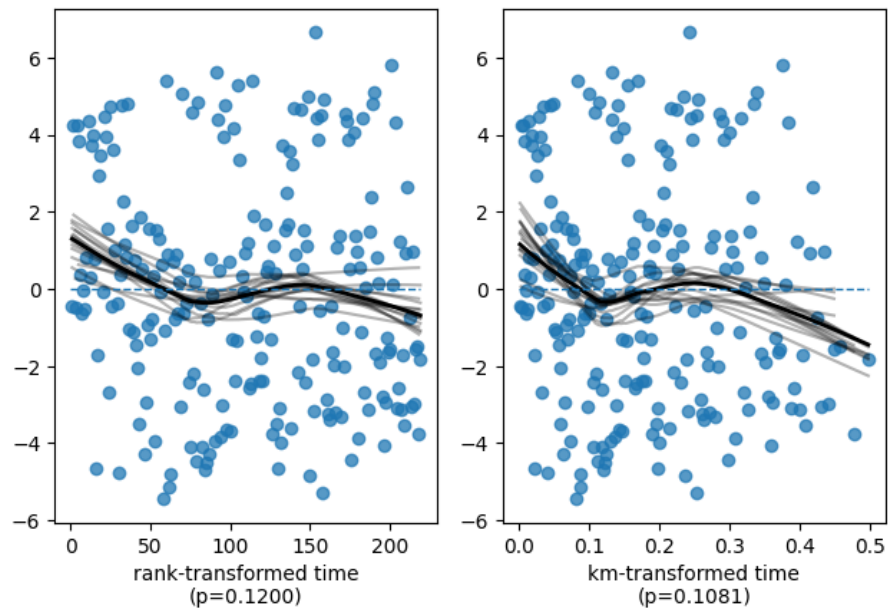
Scaled Schoenfeld residuals of 'aantalre_1.0'



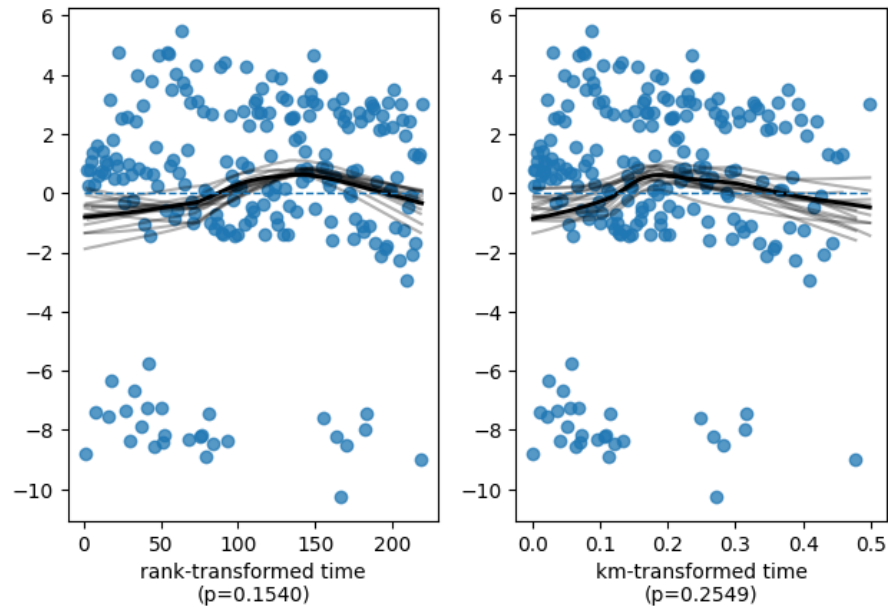
Scaled Schoenfeld residuals of 'dgf_dgf'



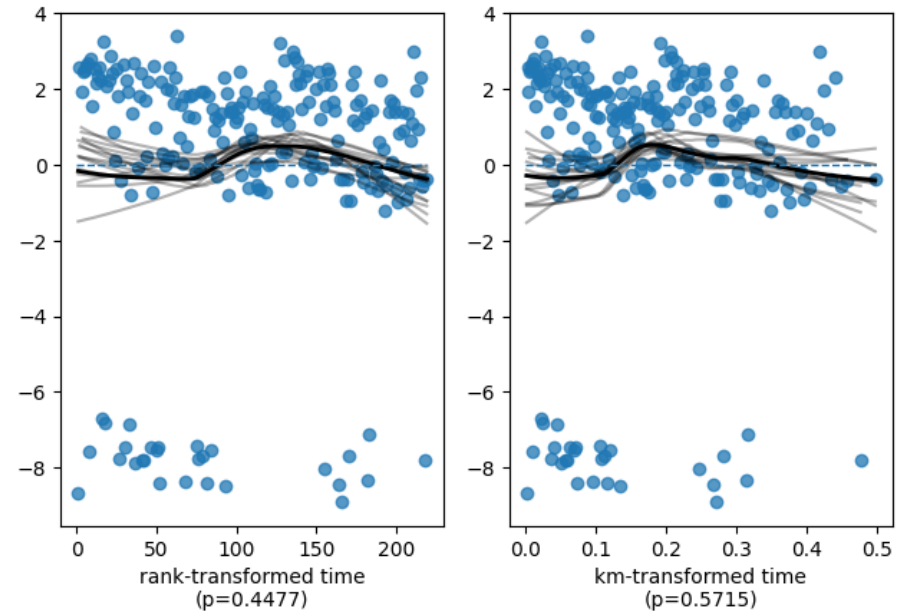
Scaled Schoenfeld residuals of 'aantalre_2.0'



Scaled Schoenfeld residuals of 'uprotein_geen'



Scaled Schoenfeld residuals of 'uprotein_sp/+'



Appendix H

covariate	Landmark Interval months	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	p	-log2(p)
acclft	LM_0-12	-0.0238	0.9765	0.0172	-0.0576	0.01	0.944	1.01	0.1669	2.5826
acclft	LM_12-24	-0.0024	0.9976	0.0137	-0.0292	0.0244	0.9712	1.0247	0.8602	0.2173
acclft	LM_24-36	0.0062	1.0063	0.0145	-0.0222	0.0347	0.9781	1.0353	0.6666	0.5851
acclft	LM_36-48	0.037	1.0377	0.0157	0.0063	0.0677	1.0063	1.07	0.0183	5.7727
acclft	LM_48-60	0.0047	1.0047	0.0197	-0.0339	0.0433	0.9666	1.0443	0.8117	0.301
acclft	LM_60-170	0.0399	1.0407	0.0104	0.0196	0.0602	1.0198	1.0621	0.0001	13.0327

Table 7 - multivariable Cox regression on accltf

covariate	Landmark Interval months	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	p	-log2(p)
aantalre_3.0	LM_0-12	2.615	13.6672	0.8035	1.0403	4.1897	2.83	66.0055	0.0011	9.783
aantalre_3.0	LM_12-24	1.8494	6.356	0.4888	0.8914	2.8074	2.4385	16.5673	0.0002	12.6591
aantalre_3.0	LM_24-36	1.358	3.8885	0.5798	0.2217	2.4943	1.2482	12.1138	0.0192	5.7056
aantalre_3.0	LM_36-48	1.1294	3.0937	0.6304	-0.1062	2.3649	0.8992	10.6434	0.0732	3.7716
aantalre_3.0	LM_48-60	0.1582	1.1714	0.773	-1.3568	1.6732	0.2575	5.3292	0.8379	0.2552
aantalre_3.0	LM_60-170	0.8547	2.3507	0.4823	-0.0905	1.7999	0.9135	6.0493	0.0763	3.7114

Table 8 - multivariable Cox regression on aantalre_3.0

Appendix I

Variable	Coef	exp(Coef)	se(Coef)	p-value	exp(-Coef)	95% CI (exp)
acclft	0.01450	1.015	0.00592	0.014	0.986	1.003–1.027
creat	-0.00037	1.000	0.00104	0.72	1.000	0.998–1.002
predias	0.00922	1.009	0.00614	0.13	0.991	0.997–1.022
prac	0.00415	1.004	0.00322	0.20	0.996	0.998–1.011
cregsh	-0.14073	0.869	0.06155	0.022	1.151	0.770–0.980
dgf_dgf	-0.39193	0.676	0.19347	0.043	1.480	0.462–0.987
aantalre_1.0	0.42900	1.536	0.18613	0.021	0.651	1.066–2.212
aantalre_2.0	0.57215	1.772	0.19913	0.0041	0.564	1.199–2.618
aantalre_3.0	1.34007	3.819	0.22914	5.0e-09	0.262	2.438–5.985
uprotein_gee n	-1.23940	0.290	0.23723	1.7e-07	3.454	0.182–0.461
uprotein_sp/ +	-0.89438	0.409	0.21999	4.8e-05	2.446	0.266–0.629

Table 1 - Fine - Gray model

Appendix J

Months	C-index	Brier Score
12	0.7107	0.0069
24	0.7325	0.0177
36	0.7032	0.0261
48	0.6214	0.034
60	0.6364	0.0398

Table 9 - cox model performance

Appendix K

